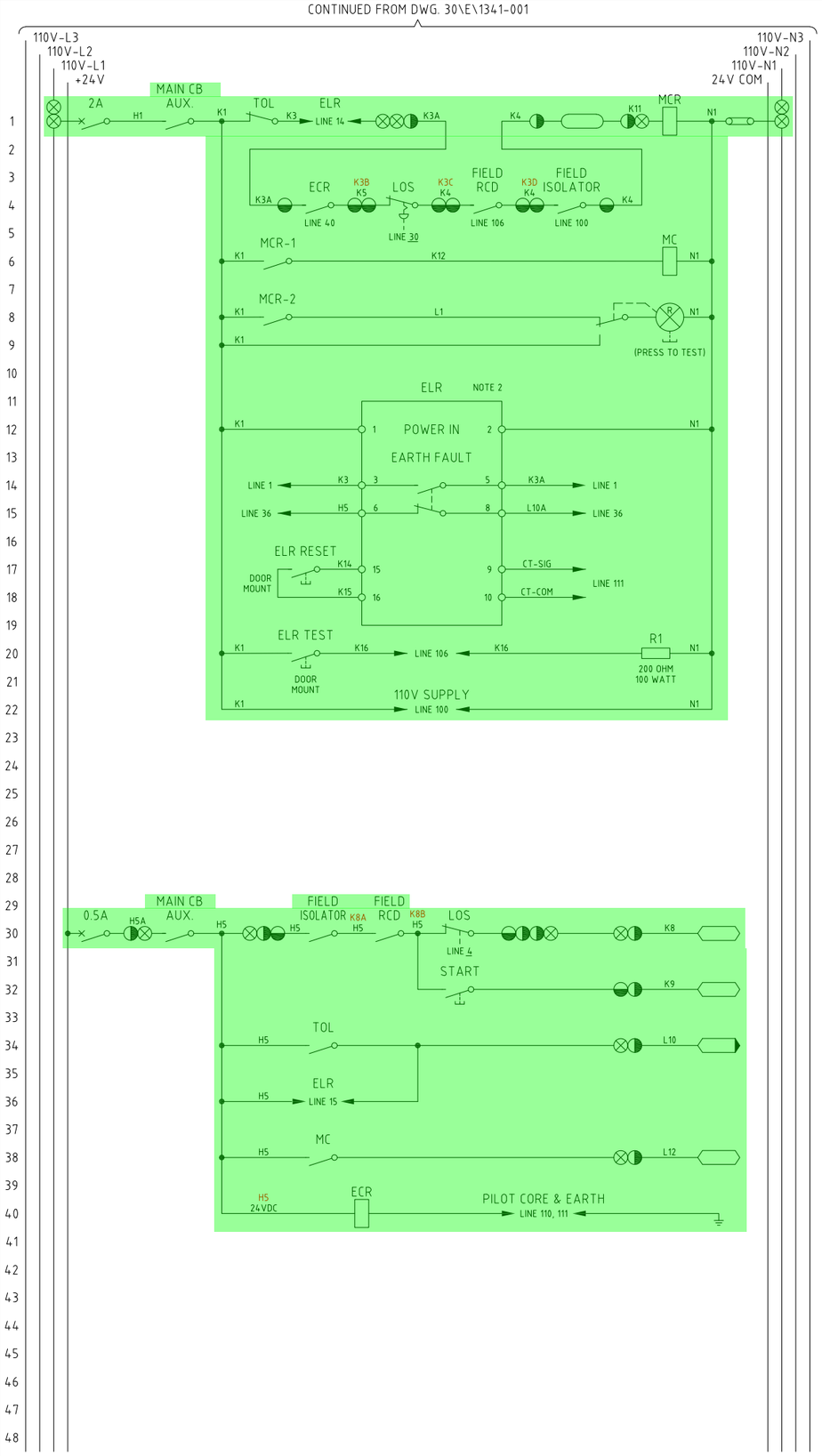


MAIN CONTACTOR RELAY
6, 8

FIELD INTERLOCKS

MAIN CONTACTOR
38, 106, 107, 108

RUN INDICATION

EARTH LEAKAGE RELAY POWER SUPPLY

ELR TEST

STOP INPUT

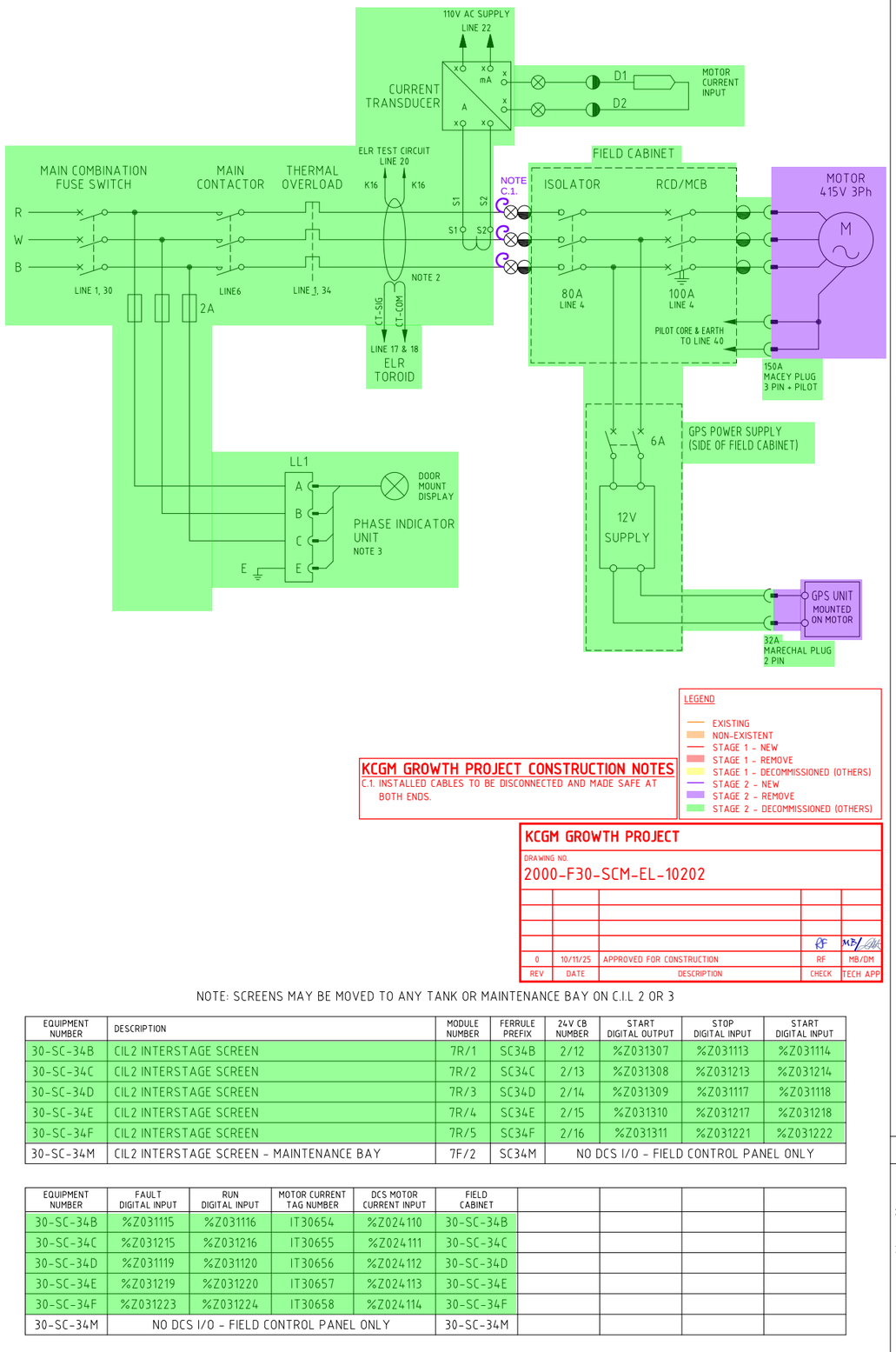
START INPUT

FAULT INPUT

RUN INPUT

EARTH CONTINUITY RELAY
4

POWER LAYOUT



LEGEND

- EXISTING
- NON-EXISTENT
- STAGE 1 - NEW
- STAGE 1 - REMOVE
- STAGE 1 - DECOMMISSIONED (OTHERS)
- STAGE 2 - NEW
- STAGE 2 - REMOVE
- STAGE 2 - DECOMMISSIONED (OTHERS)

KCGM GROWTH PROJECT CONSTRUCTION NOTES

C.1. INSTALLED CABLES TO BE DISCONNECTED AND MADE SAFE AT BOTH ENDS.

KCGM GROWTH PROJECT

DRAWING NO.
2000-F30-SCM-EL-10202

REV	DATE	DESCRIPTION	CHECK	TECH APP
0	10/11/25	APPROVED FOR CONSTRUCTION	RF	MB/DM

NOTE: SCREENS MAY BE MOVED TO ANY TANK OR MAINTENANCE BAY ON C.I.L 2 OR 3

EQUIPMENT NUMBER	DESCRIPTION	MODULE NUMBER	FERRULE PREFIX	24V CB NUMBER	START DIGITAL OUTPUT	STOP DIGITAL INPUT	START DIGITAL INPUT
30-SC-34B	CIL2 INTERSTAGE SCREEN	7R/1	SC34B	2/12	%Z031307	%Z031113	%Z031114
30-SC-34C	CIL2 INTERSTAGE SCREEN	7R/2	SC34C	2/13	%Z031308	%Z031213	%Z031214
30-SC-34D	CIL2 INTERSTAGE SCREEN	7R/3	SC34D	2/14	%Z031309	%Z031117	%Z031118
30-SC-34E	CIL2 INTERSTAGE SCREEN	7R/4	SC34E	2/15	%Z031310	%Z031217	%Z031218
30-SC-34F	CIL2 INTERSTAGE SCREEN	7R/5	SC34F	2/16	%Z031311	%Z031221	%Z031222
30-SC-34M	CIL2 INTERSTAGE SCREEN - MAINTENANCE BAY	7F/2	SC34M	NO DCS I/O - FIELD CONTROL PANEL ONLY			

EQUIPMENT NUMBER	FAULT DIGITAL INPUT	RUN DIGITAL INPUT	MOTOR CURRENT TAG NUMBER	DCS MOTOR CURRENT INPUT	FIELD CABINET				
30-SC-34B	%Z031115	%Z031116	IT30654	%Z024110	30-SC-34B				
30-SC-34C	%Z031215	%Z031216	IT30655	%Z024111	30-SC-34C				
30-SC-34D	%Z031119	%Z031120	IT30656	%Z024112	30-SC-34D				
30-SC-34E	%Z031219	%Z031220	IT30657	%Z024113	30-SC-34E				
30-SC-34F	%Z031223	%Z031224	IT30658	%Z024114	30-SC-34F				
30-SC-34M	NO DCS I/O - FIELD CONTROL PANEL ONLY				30-SC-34M				

LEGEND

TERMINALS

- MOTOR CONTROL CENTRE (MCC)
- FIELD
- PROCESS CONTROL CUBICLE
- LINE CONDITIONER
- SOFT STARTER TERMINAL
- VARIABLE SPEED DRIVE
- DISTRIBUTION BOARD
- HV MOTOR CONTROL CENTRE
- MISC. ELECTRICAL DEVICE REFER NOTES FOR SPECIFIC DESC.

DIGITAL FUNCTIONS

- DCS INPUT
- DCS INPUT WITH ALARM FUNCTION
- DCS OUTPUT

ANALOGUE FUNCTION

- 4-20mA INPUT
- 4-20mA OUTPUT

ANCILLARY EQUIPMENT

- CIRCUIT BREAKER
- COMBINATION FUSE FUSE SWITCH
- ISOLATOR
- PUSH BUTTON
- LOS (LOCK OFF STOP)
- PULL WIRE FUSE SWITCH
- NORMALLY CLOSED CONTACT
- LIMIT FUSE SWITCH
- CONTACTOR CONTACT
- NORMALLY OPEN CONTACT
- TIMER CONTACT
- FUNCTION CONTACT
  - P - PRESSURE
  - T° - TEMPERATURE
  - F - FLOW
  - F - FLOAT
- SIREN
- HORN
- RELAY
- LINK
- THERMISTOR
- INDICATION LAMP R - RED G - GREEN A - AMBER

VOLTAGES ON CONTROL BUS

+24V - 24V DC DCS SUPPLY BUS
110V-L1 - 110V AC MCC CONTROL BUS
110V-L2 - 110V AC CONDITIONED BUS
110V-L3 - 110V AC NON CONDITIONED BUS

NOTES

1. ALL WIRES EXCEPT CONTROL SUPPLIES & THOSE WITHIN FORM 4 MODULES SHALL BE FERRULED WITH PREFIX LISTED IN ABOVE TABLE.
2. EARTH LEAKAGE RELAY:-
   AMPCONTROL ELM E09792 + E08406 (DIN MOUNT)
   AMPCONTROL TOROID 60mm C18058
3. PHASE INDICATOR DETAILS: -
   LEDS - FARNELL PRODUCT No. 570035 OR EQUIVALENT
   FUSE HOLDERS - FRONT WIRED, CLIP-IN, DIN RAIL MOUNT TO COMPLY WITH BS88
   NHP CAT No. NV20FW OR EQUIVALENT

2000-F30-SCM-EL-10202_Rev0

Leach adsorption & thickening substation no. 38 415V MCC 30-SW-31 (CIL2) control schematic